

Considerations for Training an Artificial Neural Network for Particle Type Identification

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Abstract—In the nuclear sciences and radiation detection fields, the differentiation between gamma-ray and neutron interactions inside a detector volume continues to be an area of active research. Historically, the primary mechanism for conducting particle identification has been pulse shape discrimination (PSD). However, almost all variations of this technique rely on only two factors: the area of the tail and the total area of the pulse. In the last decade, the emergence of advanced machine learning techniques, most specifically artificial neural networks (ANNs), offers a unique opportunity to capitalize on the entirety of the waveform. But such techniques appear highly reliant on the quality of datasets used for training. Our research addresses this challenge to quantify the relative performances of networks trained on a variety of datasets and subjected to the same test. Furthermore, we offer an analysis of the portability of a network trained on one detector to a similar detector.

Index Terms—Machine learning, neural network, particle identification, pulse shape discrimination (PSD).

I. INTRODUCTION

A. Particle Identification

SINCE the 1950s, an important method of achieving particle identification in scintillation detectors has been pulse shape discrimination (PSD), which capitalizes on the difference in delayed fluorescence emitted by a scintillation medium excited by interaction with a gamma-ray or neutron [1]. A neutron is likely to scatter off of a hydrogen nucleus (a proton), whereas a gamma-ray is likely to give energy to an electron. An ionized proton deposits energy over a shorter track than does an electron, which allows for the occupation of a greater density of triplet states by the scintillation medium, versus a greater density of singlet states in the case of a free electron. Triplet states have a longer mean decay time, which means that the scintillation medium fluoresces over a longer period of time. This manifests in larger average tails of pulses originating from neutrons. The quotient of the integral of the tail region and the integral of the total pulse yields the pulse shape parameter (PSP), which can then be used to differentiate gamma rays and neutrons [2].

A large body of research exists for PSD. However, almost all variations of this technique rely on two factors: the area

of the tail and the total area of the pulse. Importantly, PSD methods based on pulse shape fail at low energies, where the PSP for neutron and gamma-ray pulses of the same total light output can be indistinguishable.

Another method to differentiate between gamma rays and neutrons is time-of-flight (ToF) analysis. ToF makes use of the relatively slow speeds of neutrons compared to gamma rays, which travel at the speed of light. When two detectors at different distances from a spontaneous fission source are used to measure radiation quanta, the ToF of a particle which interacts in one detector can be inferred if another particle from the same fission event is detected in the other detector [3]. However, in many measurement schemes such as those with a single detector or unknown source-detector separation, particle-type discrimination by ToF is not feasible.

B. Machine Learning

Machine learning encompasses a broad class of problems in which a computer agent executes a set of rules or algorithms to optimize a specified set of parameters. Usually, this is done for the purposes of pattern recognition, classification, or decision-making. In general, machine learning takes on three main forms: supervised, unsupervised, and reinforcement learning. Supervised learning occurs when the agent is made aware of correct answers during training such that it will continuously adjust its parameters to provide the optimal output. Conversely, unsupervised learning occurs when the agent simply groups inputs without the presence of a specified classification. Finally, reinforcement learning centers around a punishment and reward system in which the agent evolves to an optimal solution by selecting policies which optimize its reward based on its interaction with the surrounding environment.

1) *Artificial Neural Networks (ANNs)*: First theorized in the late 1940s, ANNs were designed to loosely resemble the structure of biological neurology. These mathematical models are constructed of numerous layers of “neurons” (or nodes) which receive input data, scale based on some activation function, and then transmit to subsequent layers, as demonstrated in Fig. 1.

The connections between nodes are multiplication factors known as “weights,” represented by the lines in Fig. 2. While early research proved that such ANNs could be powerful tools, it was not until the late 1980s when their full potential was realized as universal function approximators [4]. In other words, given enough data and training time, a network exists which is capable of mapping any function. As a result, ANNs have become extremely popular for pattern recognition, classification, and complex data analysis. We used the python library Keras [5] for all network development in this work.

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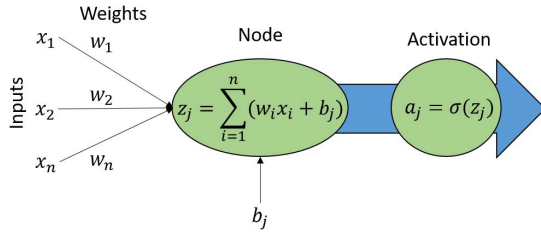


Fig. 1. Simplified flowchart of ANN information processing.

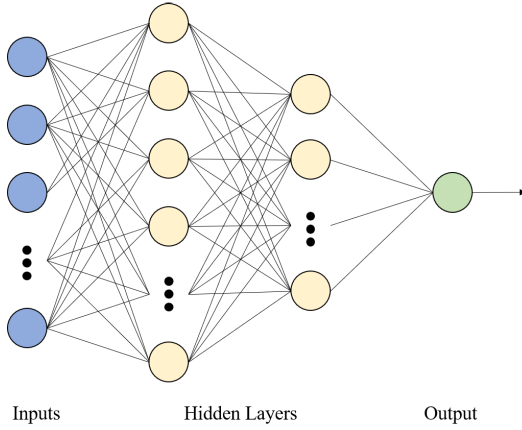


Fig. 2. Simplified diagram of neural network mapping.

During supervised learning, which we use in this research, these weights are adjusted through a process known as “back propagation” in order to minimize the collective error of the network and better match a proscribed output. Once the network is trained and weights have been sufficiently adjusted to produce adequate results, the network weights are fixed and a previously unseen test-set is introduced. The network’s performance on this final test-set is what defines the network’s overall accuracy. Thus, development of a “ground truth” test set is essential for evaluation of any supervised learning technique.

Note: A wide array of “hyperparameters” exist which modify the functionality of an ANN. Such hyperparameters include the specific structure and size of the network, learning rate, number of training iterations, and a myriad of other advanced learning techniques. For this work, we used a genetic algorithm [6] to optimize the hyperparameters, but a more deliberate examination of this choice is planned in future work [7].

2) *ANNs Used for PSD:* Machine learning algorithms have been shown to be effective for particle type identification. More specifically, ANNs were used to conduct supervised learning for mixed-field particle identification in organic scintillation detectors with an accuracy comparable to pulse shape and ToF analysis methods in 2009 by Liu *et al.* [8] and later that year by Ronchi *et al.* [9]. Further work has been done to understand the performance of ANNs at specific energy ranges [10], to handle pile-up and high count rates [11], and to compare ANN performance in different detector types, like BC-501A and BC-537 [12]. Additionally, non-ANN forms

of machine learning, such as support vector machines, have been shown to be capable of particle-type discrimination in organic scintillators [13]. However, because the creation of supervised training and testing datasets relies on *a priori* knowledge of particle type, all such efforts have to contend with contamination from misclassification, especially at low energies. At a minimum, great care must be taken to either reduce or mitigate the effect of this contamination in order to develop an accurate test for network evaluation.

C. Motivation

A review of the available literature indicates that while many articles describe methods of training a network to be very accurate, a systematic examination of the effects of various training dataset construction criteria has not been done. Therefore, the need for elaborate ToF experiment setups to create well-identified training libraries of neutrons and gamma-rays has been assumed but is not well understood.

Additionally, the stability and portability of a trained network has not been examined. We subjected a variety of trained networks to the standardized test described in Section II-A, including networks trained with detector data from scintillators at various high voltages to simulate voltage drift, and networks trained on a different detector than was used to create the test set.

II. EXPERIMENTAL PROCEDURES

In order to compare the performance of various networks, we first had to develop a standardized test dataset. To accomplish this, we created a library of classified pulses from an EJ-309 organic scintillator representing the entire energy spectrum of gamma rays and neutrons from a ^{252}Cf source classified using traditional methods. Our experimental apparatus and data processing flow were designed to minimize the number misclassifications.

A. ToF Experimental Setup

We used two detectors to capture gamma-rays and neutrons from fission events in a ^{252}Cf source. The “near detector” was a 1" lanthanum-bromide (LaBr_3) scintillator placed 0.010 ± 0.001 m from a $300\text{-}\mu\text{Ci}$ ^{252}Cf source. We selected LaBr_3 for its excellent time resolution and insensitivity to neutrons. The near detector was used to time-stamp fission events by detecting resulting gamma rays. The “far detector” was a 2" cylindrical EJ-309 suspended 0.50 ± 0.01 m from the source (see Fig. 3).

We conducted energy calibration on the far detector using ^{137}Cs , ^{60}Co , and ^{22}Na at 900 V of photomultiplier tube (PMT) bias. Additionally, we characterized the delay between near and far detector response time through a symmetrical measurement of the 511-keV annihilation photons from ^{22}Na .

More elaborate ToF experiments have been done in dedicated low-scatter spaces designed to reduce room return and chance coincidence and to allow for time separation between gamma ray and neutron interactions in the far detector [8]. Our experiment sought to approximate these conditions within the constraints of our facilities.



Fig. 3. Experimental setup. The ^{252}Cf source was touching the face of the LaBr_3 detector (the “near detector”), while the EJ-309 (the “far detector”) was suspended 50 cm away. This geometry minimized scatter into the far detector while allowing TCPH and PSD.

The CAEN V1730 digitizer with a 14-b, 2 Vpp dynamic range triggered off the far detector and recorded the event only if the near detector also had an event within $1\ \mu\text{s}$. Hundred samples were recorded for each event, at a 500-MHz sampling rate. Double pulses and pile-up events were screened by the digitizer’s firmware. Over an 18-h measurement, we captured $9.0\text{e}6$ coincident events.

B. Event Waveform Processing

For each coincident event, we calculated the PSP using (1) for each event by inverting the waveform, removing the baseline offset, and integrating the total and tail regions. The tail integral region began five samples (10 ns) after the pulse peak. The total and tail integration intervals were selected in order to maximize the separation between the time-correlated PSP (TCPSP) gamma ray and neutron regions identified in Section II-C

$$\text{PSP} = \frac{\text{tail}}{\text{total}}. \quad (1)$$

We saved waveform metadata in an array, which included pulse height, total integral, tail integral, PSP, ToF, and event identification index. We calculated ToF in reference to the timestamp of the preceding event in the near detector. The event index allowed for recall of the raw waveform for training and testing.

C. Cleaning and Labeling the Test Set

To conduct particle type identification and maximize the filtering of accidental coincidence events, we created a 3-D parameter space of PSP, ToF, and pulse height. Using this 3-D space, a series of cuts were created by requiring a minimum signal-to-noise ratio (SNR). Our choice of SNR was deliberately conservative at each step, removing only the pulses that could be confidently classified as accidental. A single waveform must clear time-correlated pulse height (TCPH), TCPSP, and a traditional pulse height versus PSP (PHvPSP) screening. Our complete procedure for particle identification is detailed below and will be referenced henceforth as TCPH-TCPSP. A similar method was used by Ronchi *et al.* [9] in 2009 to establish baseline particle-type identification of a mixed-field dataset, which was then used to compare the performance of various other particle-type classification methods.

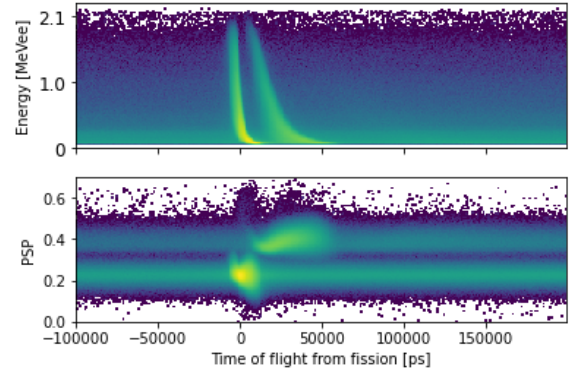


Fig. 4. TCPH (top) and PSP (bottom) histograms for the unfiltered test set data. The high-density regions in the top histogram are the gamma ray (left) and neutron (right) pulses, where the gamma-ray pulses have a shorter ToF as expected. The bottom histogram is another way of looking at the same data, where separation between gamma rays and neutrons is achieved by binning pulses by PSP and ToF. The high-density region with a short ToF and low PSP is the gamma-ray pulses, whereas the neutron pulses have a longer ToF and larger PSP.

TCPH and TCPSP plots show, respectively, the total light output and PSP of each event as a function of their ToF [14]. Fig. 4 shows the TCPH and TCPSP plots, each of which have two distinct regions: a gamma–gamma event region and a neutron–gamma event region (far detector interaction and near detector interaction, respectively). The gamma–gamma regions occur near the time origin, because these events have a very short ToF. Due to the insensitivity of LaBr_3 to neutrons, the datasets contain about half of the events that would be present using two organic liquid scintillators.

The first cut made in this step was to select bins with an SNR above chance coincidence. We defined the chance coincidence rate at each pulse height or PSP to be the average bin count for events with a ToF between $-100\,000$ and $-10\,000$ ps. Then, each TCPH and TCPSP bin was compared to determine whether it had at least twice as many events as the chance coincidence rate. All pulses in bins below this threshold were cut. We chose the SNR ratio of 2 as a conservative value that left many ($6.3\text{e}6$) events, but filtered most of the chance coincidence, as shown in Fig. 5.

Next, we identified a minimum pulse height (shown as a black horizontal line in Fig. 5) above which we could easily create two discrete continuous regions on the TCPH plot. We tentatively classified the continuous region to the right as neutrons and the continuous region to the left as gamma rays. We classified all outliers (those events in non-contiguous bins above the black line) as accidental.

For the remaining unclassified low pulse-height events, we increased the SNR threshold on the TCPSP histogram until there were two discrete continuous regions, as shown in Fig. 6. The increase in TCPSP SNR was implemented by progressively omitting the bins with the fewest counts until there was identifiable separation. This method differs from the original SNR cut where an average of PSP bin counts defined the “noise.” We used the remaining two continuous regions in the TCPSP to tentatively classify events.

To complete the cleaning of the test set, the neutron and gamma-ray events were plotted into traditional light output

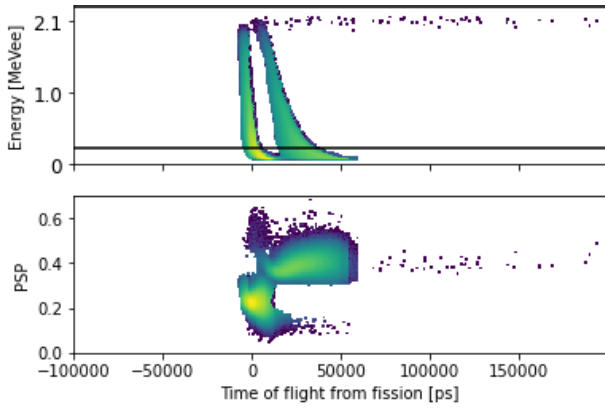


Fig. 5. TCPH (top) and TCPSP (bottom) following the implementation of a bin-by-bin SNR cut of 2 to reduce pulses arising from accidental coincidence. 6.3e6 events remain after the cuts. The black horizontal line in the top histogram represents the minimum pulse height for the classification using TCPH. Events above the line were classified according to their region, whereas events below the line were classified following an increased TCPSP SNR cut, which provided additional separation between the neutron and gamma-ray regions in the TCPSP histogram (see Fig. 6).

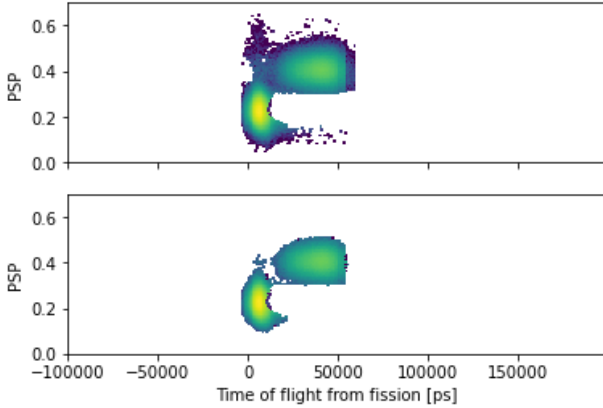


Fig. 6. TCPSP before (top) and after (bottom) increasing the SNR requirement. There are 1.7e6 events in the top plot, and 1.6e6 events remain in the bottom plot. The increased SNR cut enabled classification for the remaining pulses according to their regions. Applying this cut before the TCPH cut shown in Fig. 5 would have resulted in an unnecessary reduction in the total number of classified pulses.

versus PSP histograms, as shown in Fig. 7. Along each light output bin, we fit the PSP bin counts with a Gaussian distribution. Events that fell outside 3.5 standard deviations from the centroid were cut. This process removed less than 1e4 of the classified events.

Events that maintained consistent classification through all steps of the process described above were deemed eligible for the test set, meaning they are identified correctly to the best of our abilities. From the 18 h of data collection, we classified 1.8e6 events as neutrons, and 4.3e6 as gamma rays ranging from 50 to 2100 keVee. The events labeled as accidental events and filtered through the classification process are shown in Fig. 8 and account for nearly one-third of the initial data collected.

D. Generation of a Standardized Test Set

From the 6.1e6 events classified using the 18-h dataset, we randomly selected 1.0e6 events for the test set. Due to

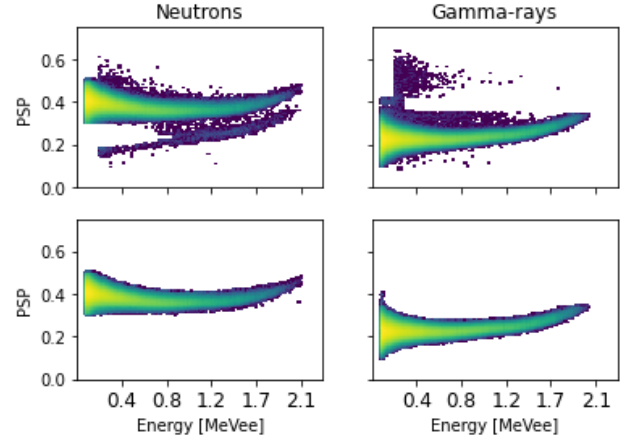


Fig. 7. Tentatively classified events plotted as PSP versus light output. We made vertical slices along the top histograms and fit Gaussian distributions to each slice. Events that were either differently classified by TCPH or TCPSP or fell outside 3.5 standard deviations from the fit were filtered. This was the final step in cleaning the baseline dataset, and removed fewer than 1e4 of the 6.1e6 remaining events.

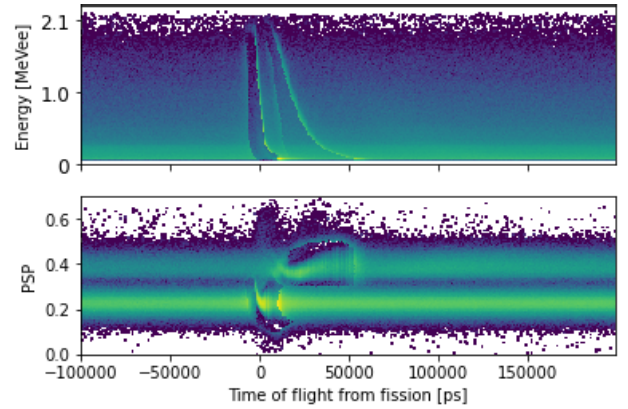


Fig. 8. 2.9e6 pulses that were not accepted and classified by the cleaning and labeling process, shown as a TCPH histogram (top), and again as a TCPSP histogram (bottom). The presence of rejected pulses in bins where the majority of pulses were successfully classified highlights the effectiveness of this iterative filtering and labeling technique.

the random selection of these events, the ratio of neutron and gamma-ray events is roughly the same 30/70 ratio expected from a ^{252}Cf source. Additionally, the events cover the full energy spectrum. As is a standard practice, the test set is never seen by a network during training and are exclusively reserved to evaluate accuracy once training is complete. In all subsequent presentations of network performance, these are the events being classified.

E. Construction of Training Datasets

We created training sets under a variety of conditions to answer the different research questions, as described in Section I-B2. Each training set was used to train ANNs consisting of similar structure and hyperparameters. In all cases, training sets were composed of a 1:1 gamma–neutron ratio to avoid network bias. Additionally, all training sets were evaluated using the previously described standardized

test set. Sections II-E1–II-E5 describe each of the training sets analyzed in this experiment.

1) *Training Using TCPH-TCPSP (Baseline)*: First, we sought to optimize the neural network hyperparameters by training networks on data from the same experiment that was used to build the test set. This training set consisted of a 1:1 ratio of gamma rays and neutrons randomly selected out of the remaining 5.1×10^6 clean events from which the standardized test set was constructed. Because this training set was comprised of data most similar to that of the testing set, we anticipated that it would achieve the most accurate results. Thus, this training set was used as the baseline for comparing other training sets. The optimization of hyperparameters is discussed in Section III.

2) *Training Using PSP Only (No ToF)*: This analysis seeks to determine the effect on the accuracy of a network trained using data from an experiment in which ToF information was not captured, for instance, in measurements involving a single organic scintillation detector or where the detector–source distance is unknown.

This dataset was compiled from the original 18-h dataset, but used only PSP discrimination to classify events. No ToF information was used, which means that the 2.9×10^6 events classified as accidental coincidence are eligible for classification. For each vertical light output slice of a light output versus PSP histogram, we fit a bimodal Gaussian to the neutron and gamma-ray PSP distributions. We removed events in a bin more than 3σ (standard deviation) away from the centroid of its particle-type Gaussian fit. This process is very effective for high-energy events, but deteriorates with lower light outputs.

At low energies, the neutron and gamma-ray regions fully overlap, and it is not possible to discriminate between the two particle types using PSP alone. In the construction of this training set, we classified any event that fell within 3σ of both the neutron and gamma-ray Gaussian fits as a gamma ray.

The impact to the training set of removing ToF information is twofold. First, chance coincidence events are included that would have been excluded with ToF information. This increases the neutron population by 290% and the gamma-ray population by 470%. Additionally, when compared to the classifications made with ToF information, approximately 2% of pulses below 70 keVee were reclassified.

3) *Training With Low-Energy Exclusion*: At higher energies, the pulse shapes of gamma-ray and neutron events are clear, even without using ToF information. However, at lower energies, these distinctions degrade and classification by PSP begins to fail. Experimenters are faced with a choice: to include the pulses in the low-energy region even though they include misclassifications (see Section II-E2) or exclude the region altogether.

The datasets in this category are exactly the same as the data from the baseline, categorized using TCPH-TCPSP (see Section II-E1), except that low-energy pulses are excluded from training. Here, only pulses with a height above 200, 400, or 800 keVee, respectively, are used.

This analysis seeks to determine the effect on the accuracy of a network trained using only pulses above various light output thresholds to avoid misclassification in the training

TABLE I
DETECTED EVENTS BY TYPE FOR VARIOUS PMT
HIGH-VOLTAGE SETTINGS

HV	Total Events	Neutrons	Gamma-rays
800	5.0×10^5	1.0×10^5	2.5×10^5
850	7.3×10^5	1.4×10^5	3.3×10^5
890	1.6×10^6	3.2×10^5	7.2×10^5
900	1.2×10^6	2.4×10^5	5.6×10^5
910	1.5×10^6	3.0×10^5	7.0×10^5
950	1.6×10^6	3.0×10^5	7.1×10^5
1000	1.8×10^6	3.1×10^5	7.8×10^5

set and to assess the predictive ability of the network in low-energy regions in which it was not trained.

4) *Training With Fluctuating High Voltage*: Scintillator response is dependent on, among other things, the high-voltage applied to the PMT. A larger applied bias results in more charge multiplication in the PMT, which can reduce energy resolution in long-duration measurements [15]. If a trained ANN is sensitive to PMT gain drift, it may degrade discrimination performance during a long measurement or necessitate the retraining of a network from day to day. Each dataset used in this analysis was taken using the same EJ-309 scintillator, LaBr₃ trigger detector, and ^{252}Cf source as described above. The bias applied to the EJ-309 PMT was ± 10 , 50, and 100 V, when compared to the test dataset experiment, which was operated at 900 V. For each high-voltage setting, data acquisition took place over the course of 3 h. The 900-V setting was also re-measured for a 3-h collection period for stability validation purposes. Table I shows the events recorded for each setting.

We conducted particle-type identification on these datasets using the same methodology described in Section II-C.

This analysis seeks to determine the effect on the accuracy of a network trained using data collected with a different PMT bias than the test set (e.g., long-duration measurements with the same detector).

5) *Detector Portability: Training on a Different Detector*: In order to begin to assess portability, we used a second EJ-309 detector of the same size to collect a separate dataset over the course of 3 h with a high voltage gain of 900 V. The two detectors were not gain-matched. We again did particle-type classification in the same manner as for the test set to attempt to achieve a minimum of misclassified events in the training sets. As with the others described above, this test set was comprised of a 1:1 ratio of gamma rays and neutrons randomly sampled from the entire set.

This analysis seeks to determine whether a trained network could be portable to another similar detector (e.g., a collection of similar detectors for which gain matching was not conducted).

III. RESULTS

We maintained consistency in network design (i.e., hyperparameters and layer configuration) for all network training. We optimized the network design using the results of a genetic algorithm that varied the hidden layer activation types, the number of hidden layers, the number of neurons per

TABLE II
ANN HYPERPARAMETERS OPTIMIZED BY GENETIC ALGORITHM

Layer	Activation Function	# nodes
Input	$1 - x/2^{14}$	100
Hidden 1	Exponential Linear Unit	180
Dropout	Rate = 0.10	
Hidden 2	Exponential Linear Unit	40
Output	Sigmoid	1

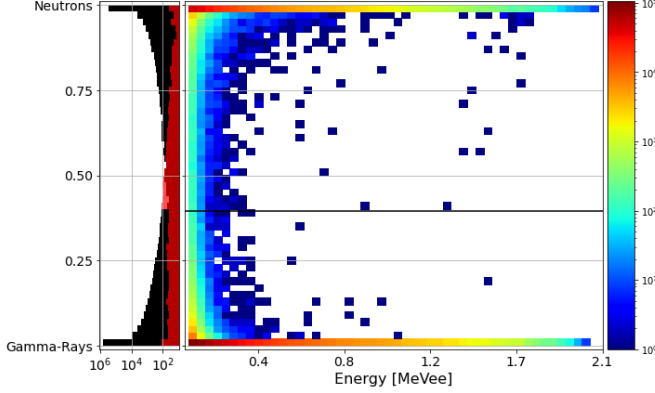


Fig. 9. Network (trained using baseline data) classification by light output of the 1.0e6 test set events. The projection to the left of the y-axis shows the distribution by classification certainty of correct classifications in black, and incorrect classifications in red, on a logarithmic scale. The black horizontal line is the line of discrimination between the neutron and gamma-ray region, chosen by an ROC curve. Events in the uppermost and bottommost row of bins are very confidently classified by the network and include 82% of total pulses.

hidden layer, and the optimizer used for training. We selected the hyperparameters (shown in Table II) that consistently provided the highest accuracy on the test set described in Section II-D.

The input provided to the trained network is 200 ns of the raw waveform (100 digitized samples at 500 MHz). The input layer inverts the signal and scales the values by the digitizer resolution (14 b) such that they are between 0 and 1.0 by dividing each pulse by its peak height. No preprocessing of the signal is required.

As a result of the sigmoid activation on the output, the network provides a result between 0 and 1 for each event. Outputs near 0 are classified as gamma rays, and values near 1 are classified as neutrons. Fig. 9 demonstrates the confidence of a network (trained using baseline data) response for the test set. The black horizontal line in Fig. 9 is the discrimination line. It was chosen by using a receiver operating characteristic (ROC) curve to maximize the network performance [16], [17]. Over 82% of the events are either in the top most, or bottom most horizontal bin (the classification is either <0.02 or >0.98).

A. TCPH-TCPSP (Baseline) Results

Fig. 10 demonstrates the performance of the network when trained under the best case scenario, where the training set is a subset of the 18-h data classified utilizing both PSP and ToF information.

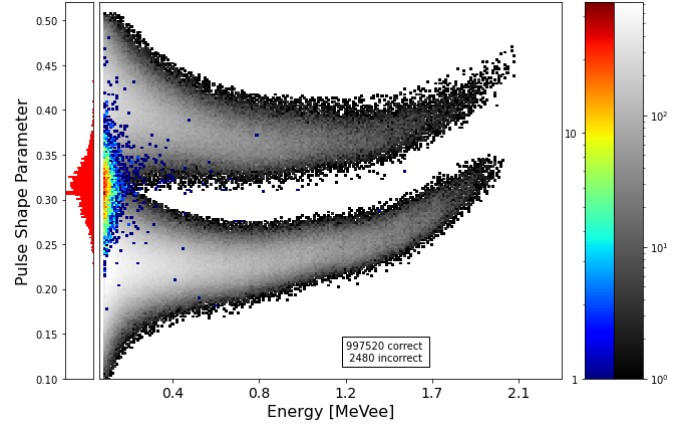


Fig. 10. Baseline network performance. Events classified correctly are represented in logarithmic gray scale density and include 99.7% of all pulses, while the misclassified events are represented by linear color density. The histogram to the left of the y-axis shows the projection of misclassified events by PSP. The distribution of misclassified events is roughly normal about the intersection of the gamma-ray and neutron regions and predominantly limited to the low-energy region.

The gray color scale represents the events that are correctly identified, while the colored bins show the density of events incorrectly classified. The neural network is able to consistently classify neutron and gamma rays with 99.7% accuracy across the full spectrum of energy deposited, in this best-case scenario. We define accuracy to be

$$\text{accuracy} = \frac{T_G + T_N}{T_G + T_N + F_G + F_N} \quad (2)$$

where T_G and T_N are correctly identified gamma rays and neutrons, and F_G and F_N are incorrectly identified gamma rays and neutrons.

All incorrectly classified events remain generally isolated to lower energies.

B. PSP Only (No ToF) Results

Network performance when trained on data categorized using only PSP is shown in Fig. 11. The network again achieved a 99.6% accuracy. There again appears to be no significant difference in performance of this training set when compared to the baseline. Both the number and distribution of incorrectly identified pulses is extremely similar.

This indicates that contamination in a training set resulting from inability to correctly classify low-energy events using only PSP does not significantly degrade network performance.

C. Low-Energy Exclusion Results

In Fig. 12, all events lower than the light output threshold (to the left of the red vertical line) are excluded from training.

In each case, as more lower-energy pulses are excluded, network accuracy marginally degrades. Energy threshold cuts of 200, 400, and 800 keV achieve an accuracy of 99.5%, 99.2%, and 98.7%, respectively. The networks appear to struggle to identify low-PSP neutron events correctly, especially when low-energy events are excluded from training.

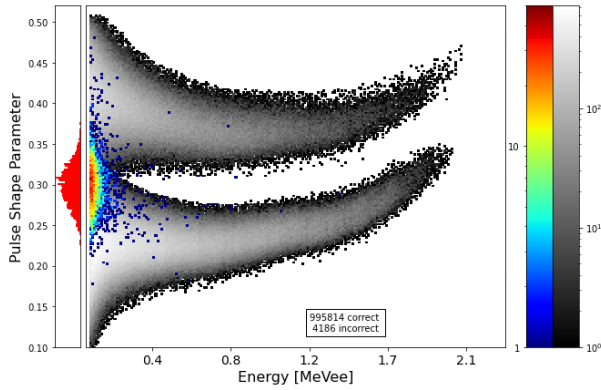


Fig. 11. Performance of network trained on events only classified by PSP. The performance of the network is similar to the baseline performance, with 99.6% of pulses correctly classified. The misclassifications are once again distributed normally by PSP around the intersection of the gamma-ray and neutron regions and limited to the low-energy domain.

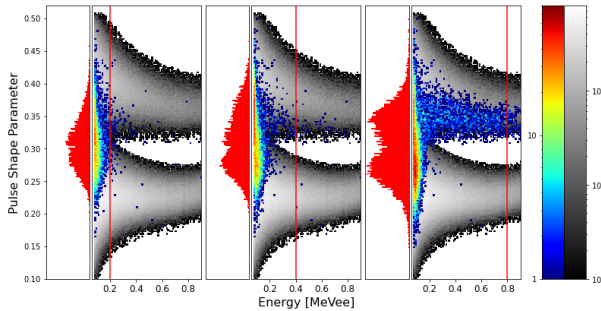


Fig. 12. Performance of networks trained only on pulses above a low-energy cut. The red vertical line indicates the lowest energy threshold at which training occurred. Pulses are still present in those regions because they were maintained in the testing dataset. Events classified correctly are represented in logarithmic gray scale density, while the events misclassified are represented by logarithmic color density. A projection of the incorrectly identified events by PSP is shown along the y-axis. Omission of low-energy pulses in training data degrades network performance, though modestly. The rates of correctly identified pulses are (from left to right) 99.5%, 99.2%, and 98.7%.

This indicates that it is preferable to include low-energy events when training an ANN to conduct PSD, even though those events will contain misclassification contamination.

D. Fluctuating High-Voltage Results

In order to assess network stability when faced with high-voltage fluctuation, 30 networks at each high-voltage setting were trained from random initial states and then assessed using data from the standard test (which was generated from data captured at 900 V). The average accuracy and variance are presented in Fig. 13.

In all cases, networks achieved reasonably accurate results. However, the 800-V networks were significantly less accurate than higher voltage training sets, only reaching an accuracy of 95%. These networks also experienced a greater amount of variance in accuracy at the conclusion of training.

Conversely, higher-voltage datasets reached an accuracy comparable to other networks analyzed in this article. Each of these networks classified more than 99% of pulses in the test set correctly. Network stability also improved, as evidenced by

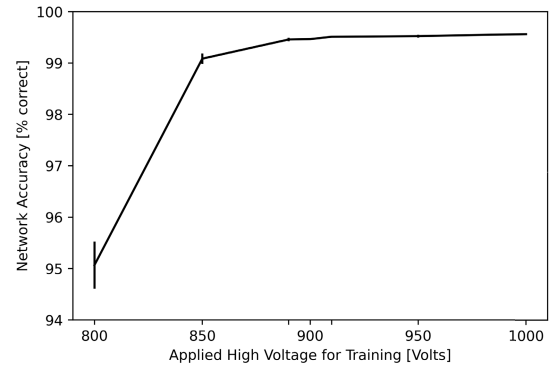


Fig. 13. Average accuracy for 30 networks trained on data collected using a high-voltage other than the 900 V used for the testing dataset. Error bars represent $2\sigma_{\bar{x}}$, or 95% confidence that a network would perform in that accuracy window. Network performance is stable from 850 to 1000 V, but degrades to 95% at 800 V.

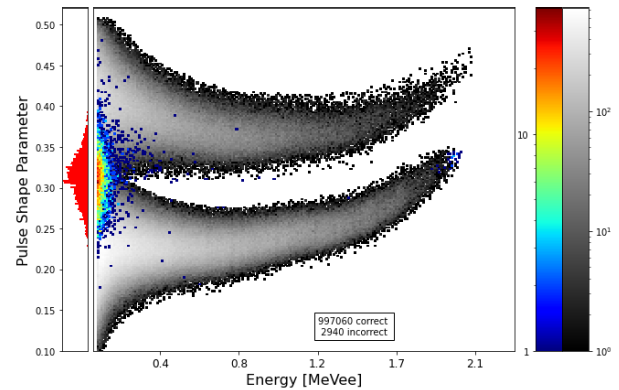


Fig. 14. Network performance when trained using data collected by a different detector than was used to create the testing dataset. Events classified correctly are represented in logarithmic gray scale density, while the events misclassified are represented by logarithmic color density. A projection of the incorrectly identified events by PSP is shown along the y-axis. The network accuracy is similar to baseline performance, with over 99.7% of pulses correctly classified. The distribution of misclassified pulses by PSP is once again normally distributed about the intersection of the gamma-ray and neutron regions and is concentrated at low energies.

the small error bars for training sets above 850 V. The worse performance of lower high-voltage datasets is expected, as the resulting pulses are smaller and, due to the finite resolution of the digitizer, carry less information.

This shows that networks can be reliable when used in a group of similar detectors that have not been gain-matched, or even when the gain applied to a PMT drifts during a long measurement.

E. Detector Portability Results

In order to establish a baseline for understanding whether a neural network is portable between detectors, we trained a network using data collected with a second EJ-309 detector and PMT assembly, identical to those used to collect the test data. The applied voltage, cabling, digitizer, and physical placement were held constant when compared to the data collected for the 1.0e6 event test set.

As shown in Fig. 14, the performance of the network remains near baseline performance (99.7% accuracy) despite training and testing on data acquired by different detectors.

This result is promising for network portability between detectors with the same physical parameters, though additional training may be beneficial for fine-tuning of a network applied to a new detector. A generalized particle-identification network could be useful in large arrays of similar detectors.

IV. CONCLUSION AND FUTURE WORK

Development of a robust training and testing sets is a crucial step in the characterization of any ANN. This axiom holds true when using ANNs for particle-type identification as well. Certainly, the utility of a network is dependent on the *a priori* classification of the training and testing sets. However, it appears that the accuracy of trained networks is stable to differences in the construction of training datasets. As long as these datasets reasonably categorize the pulses, network accuracy remains above 99%. Depending on the desired use, it is possible to exclude ToF data when developing network training without severely degrading network accuracy, thus providing the ability for rapid and accurate testing with minimal training requirements or need for lengthy re-calibration post-training as with standard PSP methods. Furthermore, ANNs appear to be robust, stable, and portable when used to conduct particle-type identification.

This research begins to articulate the impact of various training sets on overall network accuracy. However, only a limited portion of this space is explored in our current research. Future work will continue to advance along the following major lines of effort.

This article briefly addressed the issue of portability by analyzing the impact of training on a different detector. Here, we saw no significant difference in network accuracy. However, this detector was of the same type and size, a 2" cylindrical EJ-309 liquid scintillator, used in the test set. Future work will continue to investigate sensitivity of the ANN to various types of detector material and size.

Unlike supervised learning, unsupervised learning does not require the *a priori* classification of the training set. Rather, the network learns over time to group items based on characteristics it deems important. This technique has been applied by Wurtz *et al.* [18] and is an area of further research. Such a technique might allow us to avoid the challenges of labeling pulses in training data entirely.

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